

Amirjon Abdunayimov

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EDUCATION:

New York University — Tandon School of Engineering

Expected Graduation: May 2028

Bachelor of Science in Computer Science (Minor in Mathematics)

Relevant Coursework: Data Structures & Algorithms, Python Programming, Probability & Statistics, Linear Algebra, Discrete Mathematics

Academic Honors: Jack Kent Cooke Transfer Scholarship Semifinalist, Honors Scholar Designation, Phi Theta Kappa, Dean's List.

WORK EXPERIENCE:

Boardwalk Labs — Software Engineering Intern

June 2026 – Present

- Designed and deployed **Python-based LLM pipelines** and **semantic knowledge graphs**, automating AI content-processing workflows and reducing manual processing time by **40%**.
- Built a **Python/FastAPI document-ingestion API** for downstream **LLM/RAG workflows**, supporting 3 file formats (PDF, DOCX, PPTX) with text normalization, encrypted/corrupt-file handling, automated tests, and chunking-ready output.
- Led **Flutter/Dart** development across 20+ reusable components and AI learning workflows, eliminating **15+ cross-component regressions** while standardizing cross-platform UI state and theming across desktop and mobile surfaces.

Sound of Earth — DevOps & Cloud Infrastructure Intern

May 2026 – Aug 2026

- Architected containerized **CI/CD pipelines** using **Docker, GitHub Actions, and Linux**, automating multi-environment deployments, reducing production release regressions by **40%**, and saving **\$3.2K/year** in staging overhead.
- Engineered a Supabase/PostgreSQL migration for **138+ records**, restructuring Airtable data across **8 normalized tables/views** and building a PostgREST-backed migration pipeline for web and mobile clients.
- Audited production **API and infrastructure security**, identifying **29 actionable issues** across JWT, CORS, secrets management, Docker networking, CI/CD, backups, and monitoring.

Kingsborough Community College Learning Center — Computer Science & Mathematics Tutor

March 2026 – June 2026

- Mentored **50+ students across 100+ sessions** in data structures, algorithms, recursion, debugging, and mathematical problem-solving, translating complex concepts into structured implementation strategies.

PROJECTS:

sev0 — Autonomous AI Incident Investigation & Repair System | Python, Anthropic API, FastAPI, PostgreSQL, Docker, Prometheus, Loki

- Built an **autonomous AI software engineer for production incidents**, orchestrating **16 diagnostic tools** across Prometheus, Loki, Git, and Docker to trace failures to the root-cause service, file, symbol, and introducing commit.
- Designed a **4-service incident lab with 23 fault scenarios**, including **35%** adversarial and silent failures, to evaluate root-cause analysis across code, configuration, infrastructure, and business-metric regressions.
- Engineered a **sandboxed repair-and-verification pipeline** that reproduces failures before patching, runs regression checks on proposed fixes, enforces patch safety limits, and generates **draft GitHub PRs** for human review.

StateRelay — Cross-Platform Workspace State Synchronization | Go, TypeScript, Git, Linux, SQLite, mDNS, TLS

- Built a cross-platform synchronization system in **Go + TypeScript**, capturing **7+ workspace-state signals** across Git, editor, terminal, and browser context.
- Developed VS Code and browser integrations with a Go-based capture/restore pipeline, achieving complete workspace restoration across **Linux, macOS, and Windows**.
- Engineered **secure peer-to-peer transfer** using mDNS, mutual TLS, Ed25519 signatures, and SHA-256 integrity verification.

GhostMirror — Full-Stack AI Memory Engine | React, TypeScript, TailwindCSS, FastAPI, Python, SQLite, Tauri

- Built a **local-first developer activity engine** with FastAPI, SQLite, and Tauri, ingesting and indexing **500+ developer events** with **sub-100ms semantic retrieval**.
- Engineered **hybrid retrieval using SQLite FTS5 + local embeddings**, reducing search latency by **~70%** while keeping semantic computation entirely on-device.
- Developed a **React/TypeScript dashboard and 15+ REST endpoints** with event search, filtering, import/export, activity analytics, and automated testing using pytest, Vitest, and Playwright.

LEADERSHIP:

AI, Coding, and Systems (ACS) Club — Founder & President

March 2026 – June 2026

- Founded a computer science organization focused on **software engineering, AI, and systems programming**, leading technical workshops, project-based learning, and peer development sessions.

PROFESSIONAL SKILLS:

Programming Languages: Python, C++, Go, TypeScript, JavaScript, Dart, SQL

Frameworks & Data: FastAPI, React, Flutter, Node.js, PostgreSQL, SQLite

Tools & Infrastructure: Docker, GitHub Actions, Linux, Git, CI/CD, REST APIs, WebSockets, Prometheus